



Nicole Flippin
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RA-24-0068

10 CFR 50.73

March 4, 2024

U.S. Nuclear Regulatory Commission
Attention: Document Control Desk
Washington, D.C. 20555

Subject: Duke Energy Carolinas, LLC
Catawba Nuclear Station, Units 1 and 2
Docket Nos. 50-413, 50-414
Licensee Event Report (LER) 2024-001-00

Pursuant to 10 CFR 50.73(a)(1) and (d), attached is LER 2024-001-00, entitled "Condition Prohibited by Technical Specifications and Loss of Safety Function due to Failed Damper Controller for the 2A1 Emergency Diesel Generator Room Ventilation Fan."

This report is being submitted in accordance with 10 CFR 50.73(a)(2)(i)(B) and 10 CFR 50.73(a)(2)(v)(A), (B), and (D).

There are no regulatory commitments contained in this letter or its attachment.

There was no impact to the health and safety of the public.

Please direct any questions or concerns to Sherry Andrews, Sr. Engineer, at (803) 701-3424.

Sincerely,

A handwritten signature in black ink that reads "Nicole Flippin".

Nicole Flippin
Vice President, Catawba Nuclear Station

Attachment

U.S. Nuclear Regulatory Commission
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Page 2

xc (with attachment):

Laura Dudes, Region II Administrator
U.S. Nuclear Regulatory Commission
Marquis One Tower
245 Peachtree Center Ave., NE Suite 1200
Atlanta, GA 30303

Shawn Williams, Project Manager
U.S. Nuclear Regulatory Commission
11555 Rockville Pike
Mailstop O-8B1A
Rockville, MD 20852

David Rivard (without enclosure)
NRC Senior Resident Inspector



LICENSEE EVENT REPORT (LER)

(See Page 2 for required number of digits/characters for each block)
(See NUREG-1022, R.3 for instruction and guidance for completing this form
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Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch (T-6 A10M), U. S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by email to Infocollections.Resource@nrc.gov, and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk Officer for the Nuclear Regulatory Commission, 725 17th Street NW, Washington, DC 20503; email: oir_submission@omb.eop.gov. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

1. Facility Name	☒ 050 ☐ 052	2. Docket Number	3. Page
Catawba Nuclear Station, Unit 2		00414	1 OF 5

4. Title
Condition Prohibited by Technical Specifications and Loss of Safety Function due to Failed Damper Controller for the 2A1 Emergency Diesel Generator Room Ventilation Fan

5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved		
Month	Day	Year	Year	Sequential Number	Revision No.	Month	Day	Year	Facility Name	☒ 050	Docket Number
01	02	2024	2024	- 001 -	00	03	04	2024	Catawba, Unit 1		00413
									Facility Name	☐ 052	Docket Number

9. Operating Mode	10. Power Level
1	100

11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)

☒ 10 CFR Part 20	☐ 20.2203(a)(2)(vi)	☒ 10 CFR Part 50	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)	☐ 73.1200(a)
☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)	☐ 73.1200(b)
☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)	☐ 73.1200(c)
☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)	☐ 73.1200(d)
☐ 20.2203(a)(2)(i)	☒ 10 CFR Part 21	☐ 50.46(a)(3)(ii)	☒ 50.73(a)(2)(v)(A)	☒ 10 CFR Part 73	☐ 73.1200(e)
☐ 20.2203(a)(2)(ii)	☐ 21.2(c)	☐ 50.69(g)	☒ 50.73(a)(2)(v)(B)	☐ 73.77(a)(1)	☐ 73.1200(f)
☐ 20.2203(a)(2)(iii)		☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(C)	☐ 73.77(a)(2)(i)	☐ 73.1200(g)
☐ 20.2203(a)(2)(iv)		☒ 50.73(a)(2)(i)(B)	☒ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(ii)	☐ 73.1200(h)
☐ 20.2203(a)(2)(v)		☐ 50.73(a)(2)(i)(C)	☐ 50.73(a)(2)(vii)		

☐ OTHER (Specify here, in abstract, or NRC 366A).

12. Licensee Contact for this LER

Licensee Contact	Phone Number (Include area code)
Sherry Andrews, Senior Engineer, Regulatory Affairs	803-701-3424

13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable to IRIS	Cause	System	Component	Manufacturer	Reportable to IRIS
B	VJ	TC	L321	YES					

14. Supplemental Report Expected

☒ No	☐ Yes (If yes, complete 15. Expected Submission Date)	15. Expected Submission Date	Month	Day	Year

16. Abstract (Limit to 1326 spaces, i.e., approximately 13 single-spaced typewritten lines)

On January 2, 2024, the 2A Emergency Diesel Generator (DG) was unloaded and secured prior to the completion of a periodic 5-hour run due to rising temperatures and air quality conditions in the 2A DG room. It was determined that the 2A1 Emergency Ventilation Fan Damper Controller was failed and did not allow the normal amount of outside air to circulate into the room. Additionally, the 2A2 Emergency Ventilation Fan was out of service in support of maintenance activities. As a result, 2A DG room temperatures increased to an off-normal level less than the alarm setpoint. The 2A DG was declared inoperable until the degraded damper controller was repaired.

Further investigation revealed Catawba Nuclear Station had evidence the condition of the failed 2A1 Damper Controller, combined with the out of service 2A2 Ventilation Fan, had existed since December 5, 2023. A review performed for the timeframe between December 5, 2023, and January 2, 2024, identified the 2A DG VD system was non-functional during this time, resulting in the 2A DG being inoperable. This resulted in instances which represented conditions prohibited by Technical Specifications, and instances of a loss of safety function. There was no impact to the health and safety of the public.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
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1. FACILITY NAME Catawba Nuclear Station, Units 1 and 2	☒ 050	2. DOCKET NUMBER 00413, 00414	3. LER NUMBER		
	☐ 052		YEAR 2024	SEQUENTIAL NUMBER - 001	REV NO. - 00

NARRATIVE**BACKGROUND**

The following information is provided to assist readers in understanding the event described in this LER. Applicable Energy Industry Identification System [EII] and component codes are enclosed within brackets. Catawba's unique system and component identifiers are contained within parentheses.

This event is being reported under 10 CFR 50.73(a)(2)(i)(B) for condition prohibited by Technical Specifications (TS) and 10 CFR 50.73(a)(2)(v)(A), (B), and (D) for an event or condition that could have prevented the fulfillment of a Safety Function.

The onsite standby power source for each 4160 volt Engineered Safety Features (ESF) bus at Catawba is a dedicated Diesel Generator (DG)[EK]. For each unit, DGs A and B are dedicated to ESF buses ETA and ETB, respectively. Each DG starts automatically on a Safety Injection (SI) signal (i.e., low pressurizer pressure or high containment pressure) or on an ESF bus degraded voltage or undervoltage signal. After the DG has started, it will automatically tie to its respective bus after offsite power is tripped as a consequence of ESF bus undervoltage or degraded voltage, independent of or coincident with an SI signal. The DGs will also start and operate in the standby mode without tying to the ESF bus on an SI signal alone. Following the trip of offsite power, a sequencer strips loads from the ESF bus. When the DG is tied to the ESF bus, loads are then sequentially connected to its respective ESF bus by the automatic load sequencer. In the event of a loss of preferred power, the ESF electrical loads are automatically connected to the DGs in sufficient time to provide for safe reactor shutdown and to mitigate the consequences of a Design Basis Accident such as a Loss of Coolant Accident (LOCA).

There are also provisions to accommodate the connecting of the Emergency Supplemental Power Source (ESPS) to one train of either unit's Class 1E AC Distribution System (ESF). The ESPS consists of two 50% capacity non-safety related commercial grade DGs. Manual actions are required to align the ESPS to the station and only one of the station's four onsite Class 1E Distribution System trains can be supplied by the ESPS at any given time. The ESPS is made available to support extended Completion Times in the event of an inoperable DG as well as a defense-in-depth source of AC power to mitigate a station blackout event. The ESPS would remain disconnected from the Class 1E AC Distribution System unless required for supplemental power to one of the four 4.16 kV ESF buses.

The primary purpose of the Diesel Building Ventilation System (VD)[VJ] is to maintain the temperature within the minimum and maximum limits for operation of equipment and personnel access for inspection, testing, and maintenance. The ESF Actuation System actuates the Diesel Building Ventilation System to maintain proper ventilation of the Diesel Building for equipment operation. The Diesel Generator Emergency Ventilation portion of the VD System is required to maintain the Diesel Building temperature between 60 and 120-degrees Fahrenheit during any Design Basis Event (such as a LOCA) coincident with a Loss of Offsite Power (LOOP). Emergency Ventilation Fans 2A1 and 2A2 automatically start when the 2A DG starts.

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	☐ 052		<table border="1"><thead><tr><th>YEAR</th><th>SEQUENTIAL NUMBER</th><th>REV NO.</th></tr></thead><tbody><tr><td>2024</td><td>- 001</td><td>- 00</td></tr></tbody></table>	YEAR	SEQUENTIAL NUMBER	REV NO.	2024
YEAR	SEQUENTIAL NUMBER	REV NO.					
2024	- 001	- 00					

NARRATIVE

Catawba TS 3.8.1 governs the DGs for each unit that is in Modes 1, 2, 3, and 4. Limiting Condition for Operation (LCO) 3.8.1(b) requires two operable DGs capable of supplying the Onsite Essential Auxiliary Power Systems. With one LCO 3.8.1(b) DG inoperable, the inoperable DG must be restored to operable status within 72 hours per Required Action B.6. LCO 3.8.1(d) requires two DGs from the opposite unit necessary to supply power to the shared systems and the Nuclear Service Water System (NSWS) pump(s). With one LCO 3.8.1 (d) DG inoperable, the inoperable DG must be restored to operable status within 72 hours per Required Action D.6. Additionally, both Conditions B and D contain 4-hour actions to declare supported systems inoperable upon discovery of the inoperability of redundant required features concurrent with DG inoperability, Conditions B.3 and D.3 respectively.

If an inoperable DG is not restored to operable status within 72 hours, the unit must be placed in Mode 3 within 6 hours and in Mode 5 within 36 hours per LCO 3.8.1 Required Actions I.1 and I.2.

In order to extend the Completion Time for an inoperable DG from 72 hours to 14 days, it is necessary to ensure the availability of the ESPS within 1 hour of entry into TS 3.8.1 LCO and every 12 hours thereafter.

Catawba TS 3.7.10 governs the Control Room Area Ventilation System (CRAVS). LCO 3.7.10 requires two CRAVS trains operable in Modes 1 through 6. Condition F states to enter LCO 3.0.3 immediately when two trains of CRAVS are inoperable in Mode 1, 2, 3, or 4 for reasons other than Condition B.

Catawba TS 3.7.11 governs the Control Room Area Chilled Water System (CRACWS). LCO 3.7.11 requires two CRACWS trains operable in Modes 1 through 6. Condition E states to enter LCO 3.0.3 immediately when two trains of CRACWS are inoperable in Mode 1, 2, 3, or 4.

At the time this condition was identified, Catawba Units 1 and 2 were operating in Mode 1 at approximately 100 percent power. No additional significant structures, systems or components were out of service at the time of this event that contributed to this event.

EVENT DESCRIPTION

On January 2, 2024, at approximately 1440, the 2A Emergency Diesel Generator (DG) was unloaded and secured prior to the completion of a periodic 5-hour run due to rising temperatures and air quality conditions in the room. It was determined the 2A1 Emergency Ventilation Fan Damper Controller was failed and did not allow the normal amount of outside air to circulate into the room. Additionally, the 2A2 Emergency Ventilation Fan was out of service in support of maintenance activities. As a result, 2A DG room temperatures increased to an off-normal level less than the alarm setpoint. The DG room ventilation systems are design features and thus their functionality impacts the operability of the DG system. The 2A DG was declared inoperable until the failed damper controller was repaired.

A review of historical room temperatures from the operator aid computer (OAC) for the 2A DG revealed Catawba Nuclear Station had evidence the failed condition with the 2A1 Emergency Ventilation Fan Damper Controller, combined with the out of service 2A2 Emergency Ventilation Fan, had existed since December 5, 2023. Therefore, the 2A DG was inoperable between December 5, 2023, and January 2, 2024.

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CONTINUATION SHEET**

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NARRATIVE

A review performed for this timeframe identified that the 2A DG exceeded Required Actions and allowable Completion Times of TS 3.8.1, Condition B.6 for Unit 2, Condition D.6 for Unit 1, and Conditions I.1 and I.2 for both Units 1 and 2. This represents an "operation or condition prohibited by Technical Specifications" reportable under 10 CFR 50.73(a)(2)(i)(B) for both Units 1 and 2.

Additionally, instances were identified that represent a "condition that could have prevented fulfillment of a safety function" reportable under 10 CFR 50.73(a)(2)(v). At the time of discovery on January 2, 2024, the 2B DG and all shared and Unit 2 supported systems were operable.

On December 19, 2023, from 0302 until 0445, and 0904 until 1112, the 2B DG was declared inoperable in support of planned testing. Therefore, this represents a "condition that could have prevented fulfillment of a safety function" reportable under 10 CFR 50.73(a)(2)(v) for Unit 2.

From December 17, 2023, at 1543, until December 19, 2023, at 2057, the B Train CRAVS and B Train CRACWS shared systems were declared inoperable for planned maintenance. LCO 3.8.1, Condition D, Required Action D.3, directs declaring shared systems supported by the inoperable 2A DG inoperable 4 hours from discovery of inoperability of redundant required features. Therefore, A Train CRAVS and CRACWS were inoperable on December 17 at 1943. LCOs 3.7.10 (CRAVS) Condition F and 3.7.11 (CRACWS) Condition E require immediate entry into LCO 3.0.3 for two inoperable trains. This condition represents a "condition that could have prevented fulfillment of a safety function" reportable under 10 CFR 50.73(a)(2)(v), and an "operation or condition prohibited by Technical Specifications" reportable under 10 CFR 50.73(a)(2)(i)(B) for Unit 1 and Unit 2.

CAUSAL FACTORS

The cause of the degraded 2A1 Emergency Ventilation Fan Damper Controller was determined to be faulty internal wiring. A failure analysis, which included a detailed disassembly and inspection, was performed on the controller. The inspection identified internal wiring and soldering quality issues associated with the temperature transmitter for the 2A1 Emergency Ventilation Fan Damper Controller.

CORRECTIVE ACTIONS

1. Repaired and returned the 2A1 damper controller to service.
2. Returned the 2A2 VD fan to service.
3. Evaluate improvements to the commercial grade dedication process for the damper controllers.



LICENSEE EVENT REPORT (LER) CONTINUATION SHEET

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NARRATIVE

SAFETY ANALYSIS

This event did not result in a plant transient, radiological release, or other challenge to normal plant operation to cause any impacts to the public health and safety. Catawba Units 1 and 2 have a considerable level of defense-in-depth besides having two DG trains. The ESPS provides a means to restore AC power to the 2A Essential Bus (2ETA) if the 2A DG failed to run. Similarly, another recovery action is the ability to align offsite power from Unit 1 if available via the Shared Auxiliary Transformer SATA. The Safe-Shutdown Facility (SSF) also provides an alternative means of maintaining safe shutdown conditions with the SSF Diesel Generator powering the Standby Makeup Pump for reactor coolant pump (NCP) seal cooling and instrumentation and controls for operation of Turbine-Driven Auxiliary Feedwater Pump. The final level of safety is the ability to use FLEX equipment and strategies to provide alternative core cooling to prevent core damage.

With these factors taken together, there was no impact to the health and safety of the public.

ADDITIONAL INFORMATION

Previous Similar Occurrences:

A review of License Event Reports from the past 3 years did not identify any similar occurrences.